RESEARCH PROPOSAL

Toward a Topological Framework for Cognition

Memory, Learning, and Concept Formation as Persistent Structure in Dynamical State Spaces

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Timeline: 6 Months · Goal: Research Paper

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1. Motivation and Problem Statement

Modern artificial intelligence systems achieve remarkable performance in pattern recognition, language generation, and sequential decision-making. Yet there exists no widely accepted mathematical framework describing what memory, conceptual understanding, and learning actually *are* as structural phenomena. The dominant paradigm treats intelligence as an optimization problem: minimize a loss function over a parameter space using gradient descent, represent knowledge as points in high-dimensional vector spaces, and predict future states from past observations.

This paradigm is built almost entirely on two mathematical foundations: **linear algebra** (vectors, matrices, tensor operations, projections) and **probability theory** (likelihoods, conditional distributions, next-token prediction). These tools are powerful for computation, but they provide no native language for persistence, structural organization, relational invariance, or continuity of experience. A transformer can predict the next token with extraordinary accuracy, but it has no formal description of what a concept is, what it means for memory to persist, or what distinguishes genuine understanding from sophisticated interpolation.

This proposal advances a specific hypothesis: that core cognitive phenomena—memory persistence, concept formation, relational understanding, learning, and continuous integrated experience—may be better described using **topology and dynamical systems** as a complementary structural layer. The claim is not that linear algebra and probability are wrong. It is that they are incomplete. They describe coordinates and likelihoods; topology describes structure and persistence.

The central question is whether topology provides a useful mathematical language for describing cognition—one that captures what existing frameworks miss.

2. What Current Models Do and What They Miss

2.1 The Dominant Paradigm

Current leading AI architectures can be organized by what mathematical objects they operate over and what they optimize:

Architecture	Core Abstraction	Optimizes
Transformers	Token sequences, attention-weighted context vectors	Next-token log-likelihood
State-Space Models (S4, Mamba)	Latent state vectors evolving via learned dynamics	Sequence prediction loss
World Models	Latent state z, transition f(z, a), observation g(z)	Reconstruction + reward
Representation Learning	Embedding vectors in metric spaces	Contrastive / reconstruction loss
Graph Neural Networks	Node and edge feature vectors on graphs	Node/graph classification loss

Table 1. Current AI architectures and their mathematical foundations.

In every case, the core mathematical objects are vectors, the core operation is matrix multiplication or attention, and the core objective is minimizing a scalar loss. Knowledge is a point in space. Learning is moving that point. Prediction is computing a conditional distribution.

2.2 The Structural Gap

These frameworks answer: *what vector represents this input?* They do not naturally answer: *what is the structure of knowledge?* Consider the following cognitive phenomena and how the current paradigm handles them:

Phenomenon	Current Treatment	What Is Missing
Memory	Stored vectors; context window; external database lookup	No account of persistence, stability under perturbation, or structural organization of what is remembered
Concepts	Clusters or regions in embedding space defined by distance	No formal description of what makes a concept stable, how it relates to others structurally, or when it has formed
Learning	Parameter update via gradient descent	No description of structural change; only numeric change in coordinates
Understanding	Embedding similarity; benchmark accuracy	No distinction between memorization and genuine structural comprehension
Reasoning	Chain-of-thought prompting; sequential token generation	No formal account of relational traversal through structured knowledge

Table 2. Cognitive phenomena and what the dominant paradigm misses.

The unifying observation is that current models represent knowledge as **geometry** (distances between points) but have no native representation of **topology** (persistent structure, connectivity, holes, relational organization). Geometry tells you how far apart two things are. Topology tells you how they are connected.

2.3 The Specific Distinction from World Models and State-Space Models

A likely objection is that world models and state-space models already use dynamical systems. This is true. They model latent state transitions: $z_{t+1} = f(z_t, a_t)$. But they model *what state comes next*. This proposal asks a different question: what is the *structure* of the state space itself, how does that structure form and stabilize during learning, and do the persistent structural features of that space correspond to what we recognize as memory and concepts?

World models learn trajectories through a space. This project studies the topology of the space those trajectories inhabit.

3. Core Hypothesis

We hypothesize that core cognitive phenomena—memory persistence, concept formation, relational understanding, and continuous experiential integration—can be modeled as **topological structures evolving within dynamical state spaces**. If correct, topology provides a structural framework complementary to the linear algebraic and probabilistic foundations of current AI, capturing properties of cognition that those foundations do not naturally express.

The hypothesis decomposes into five testable sub-claims:

Sub-Hypothesis 1: Memory as Persistent Structure

Memory corresponds to topological features in a cognitive state space that persist under perturbation. A system that remembers is one whose state space contains structures (connected components, loops, attractors) that are stable across time and robust to noise. Forgetting corresponds to the collapse or pruning of such structures. Persistent homology provides a natural formalism: features with long persistence lifetimes correspond to stable memories; short-lived features correspond to transient sensory impressions.

Sub-Hypothesis 2: Concepts as Stable Topological Regions

A concept is not a point in embedding space but a structured region or manifold with characteristic topological properties—boundary, internal connectivity, and relational links to other concept regions. Concept formation is the emergence and stabilization of such regions during learning. A concept has *formed* when a region with stable topological invariants (Betti numbers, homology classes) has crystallized in representation space.

Sub-Hypothesis 3: Learning as Structural Deformation

Learning is not merely coordinate change (parameter update) but topological reorganization of the state space. Early learning creates new connected components and bridges between regions. Deep learning refines the internal structure of existing regions. Overfitting corresponds to excessive structural rigidity; generalization corresponds to the right level of topological complexity for the data.

Sub-Hypothesis 4: Understanding as Structure-Preserving Mapping

Understanding a domain means having an internal topological structure that is structurally consistent with the domain's actual organization. A system *understands* when there exists a continuous or structure-preserving map between its internal concept space and the relational structure of the external domain. This distinguishes genuine comprehension from statistical interpolation.

Sub-Hypothesis 5: Continuous Experience as Recurrent Global State Integration

The continuous loop of moment-to-moment experience (what we term the *Recursive Integrative Experience Loop*) corresponds to a recurrent dynamical process that continuously updates a globally connected state, integrating sensory input, memory recall, and motor output into a coherent present. The topological requirement is global connectivity: the experience loop must maintain a connected state space at every timestep.

4. Proposed Mathematical Objects

The framework requires defining specific mathematical objects. The following are preliminary and subject to refinement during the research:

Object	Definition	Cognitive Role	Topological Measure
Cognitive State Space S	High-dimensional topological space; evolves under dynamics	Contains all structure representing knowledge and experience	Dimension, global connectivity, homology
Memory Structure M ⊆ S	Persistent subcomplex; attractor or stable region	Stored information that survives perturbation	Persistence lifetime, Betti numbers
Concept Region C ⊆ S	Connected region with stable boundary and internal structure	Abstraction over instances; categorical knowledge	Boundary homology, internal Betti numbers
Learning Operator L	Map L: S → S′ that deforms the topology of the state space	Structural reorganization during training	Change in persistence diagrams pre/post L
Understanding Map U	Continuous or structure-preserving map between concept spaces	Structural correspondence: internal model ↔ external domain	Degree of homomorphism; structural distortion
Experience Loop E(t)	Recurrent dynamical process; continuous global state update	Moment-to-moment integrated cognition	Global connectedness at each timestep t

Table 3. Proposed mathematical objects, their cognitive roles, and topological measures.

These definitions are the starting point, not the conclusion. A primary goal of the research is to determine which of these objects are well-defined and which require reformulation.

5. Testable Hypotheses Leading to Paper 1

The first paper will not attempt the full cognitive framework. It will test a tightly scoped subset of claims that are experimentally accessible within six months. Specifically:

Hypothesis H1: Persistent Topological Features Emerge During Training

When a neural network learns to classify inputs into distinct categories, its internal representation space develops topological features (connected components, loops) whose persistence lifetimes increase monotonically as training progresses and classification accuracy improves.

Test: Train a standard classifier (e.g., a small CNN on MNIST or CIFAR-10). Extract hidden layer activations at regular intervals during training. Compute persistence diagrams of the point clouds formed by these activations. Track persistence lifetimes of the most prominent features across epochs. The hypothesis predicts that persistence lifetimes grow and stabilize as the network converges.

Hypothesis H2: Topological Structure Correlates with Learned Categories

The number of persistent connected components in a trained network's representation space corresponds to the number of learned categories. Each persistent component contains points from predominantly one class.

Test: After training, compute the 0-dimensional persistence diagram (connected components) of the final hidden layer activations. Count the number of components with persistence lifetime above a threshold. Compare to the number of classes. Measure class purity within each component. The hypothesis predicts a close correspondence.

Hypothesis H3: Overfitting Produces Excess Topological Complexity

A network that overfits its training data develops more topological features (higher Betti numbers, more persistent loops and voids) than a well-generalizing network trained on the same data. Overfitting corresponds to unnecessary structural complexity in representation space.

Test: Train two networks: one regularized (dropout, weight decay) and one unregularized. Compare persistence diagrams and Betti numbers of both at convergence. The hypothesis predicts that the overfitting network has strictly higher topological complexity.

Hypothesis H4: Learning Is Measurable as Topological Change

The Wasserstein distance between persistence diagrams at successive training epochs provides a meaningful measure of structural learning rate. This topological learning rate correlates with but is not identical to the loss curve—structural reorganization may occur in bursts that do not correspond to smooth loss descent.

Test: Compute Wasserstein distances between consecutive persistence diagrams throughout training. Plot topological learning rate alongside the loss curve. Look for phase transitions: epochs where topological structure changes sharply even when loss changes gradually (or vice versa).

6. From Paper 1 to a Novel Architecture

If the hypotheses in Section 5 are confirmed—if topology does capture structural properties of learning that loss curves miss—then the results become the empirical foundation for a much larger claim: that cognition itself can be modeled as structure formation in topological state spaces.

6.1 The Architectural Vision

The BrAIn architecture is a cognitive operating system in which a model does not merely process sequences or predict tokens but maintains a persistent, structured, evolving state space that constitutes its memory, knowledge, and understanding. The architecture includes:

- **A Cognitive Loop** that continuously integrates sensory input, recalls from structured memory, reasons over a state graph, gates proposed actions through a safety policy engine, executes, and remembers the outcomes.
- **Three-Layer Memory:** episodic memory (timestamped traces of events, actions, outcomes), semantic memory (a knowledge graph of concepts and relations), and procedural memory (learned workflows and refined pathways).
- **A Body Map** that incorporates hardware devices not as external tools but as extensions of the model's own state space—expanding the state graph when new devices are connected, forming proprioceptive models of each device's behavior.
- **A Safety Pipeline (KIRA)** that evaluates every proposed action through eight formal stages before execution, running in a hardware-isolated domain that the model cannot address or modify.

6.2 Why Topology Is Essential to This Architecture

The BrAIn architecture requires mathematical tools that current AI does not provide:

1. **Memory persistence.** BrAIn's episodic, semantic, and procedural memories must persist and remain structurally stable across millions of cognitive loop iterations. Persistent homology provides a formal measure of which structures are robust and which are transient.
2. **Concept formation detection.** BrAIn needs to know when it has learned a new concept versus when it has merely seen new data. Monitoring the Betti numbers and persistence diagrams of its semantic graph provides a structural signal independent of any specific loss function.
3. **Structural learning rate.** BrAIn must distinguish productive learning (structural reorganization that improves capability) from drift (structural change that degrades capability). The topological learning rate from Hypothesis H4 provides this.
4. **Body map coherence.** When BrAIn incorporates a new device, the expanded state space must remain globally connected. Monitoring H_0 (connected components) during device incorporation ensures coherence.
5. **Understanding verification.** When BrAIn claims to understand a domain, there should be a verifiable structural correspondence between its internal concept space and the domain's relational structure. Structure-preserving maps between topological spaces formalize this.

6.3 Paper 2: The Groundbreaking Paper

If Paper 1 establishes the empirical basis, Paper 2 synthesizes the full framework:

> *Working Title: "Topological Cognition: A Structural Framework for Memory, Learning, and Understanding in Dynamical State Spaces"*

This paper will present the formal mathematical framework (the objects in Section 4, refined by experimental results), the experimental evidence from Paper 1, a toy implementation demonstrating topology-aware learning (a small network trained with a topological regularization term that penalizes unnecessary structural complexity), and the connection to the BrAIn cognitive architecture as a concrete system where the framework is applied.

Possible venues: NeurIPS, ICML, ICLR (main conference or workshop), SIAM Journal on Applied Mathematics, Journal of Machine Learning Research, or Physical Review E if the physics connection is emphasized.

7. Methods and Computational Tools

7.1 Mathematical Methods

Method	Application	Key Reference
Persistent Homology	Measuring structural features and their lifetimes in representation spaces	Edelsbrunner & Harer, Computational Topology
Persistence Diagrams	Visual and quantitative summary of topological features at multiple scales	Carlsson, Topology and Data (2009)
Wasserstein Distance	Measuring structural distance between persistence diagrams across training epochs	Cohen-Steiner et al., Stability of Persistence Diagrams
Betti Numbers	Counting connected components, loops, voids in representation space	Standard algebraic topology
Dynamical Systems Theory	Modeling the evolution of state spaces under learning dynamics	Kelso, Dynamic Patterns
Attractor Analysis	Identifying stable memory structures as attractors in state space	Hopfield, Neural Networks (1982)

Table 4. Mathematical methods and their applications in this research.

7.2 Computational Tools

- **Ripser / Gudhi / Giotto-TDA:** libraries for computing persistent homology and persistence diagrams efficiently.
- **PyTorch:** training neural networks and extracting intermediate representations.
- **UMAP / t-SNE:** visualizing high-dimensional representation spaces for qualitative analysis.
- **NetworkX / graph-tool:** analyzing the topology of semantic knowledge graphs.
- **SciPy / NumPy:** numerical computation, distance metrics, statistical analysis.

7.3 Experimental Systems

Experiments will use simple, well-understood learning systems where structural claims can be verified:

1. Image classifiers (CNN on MNIST, CIFAR-10) for Hypotheses H1–H3.
2. Autoencoders trained on structured data for studying representation topology.
3. Hopfield-style attractor networks as toy models of memory persistence.
4. Simple recurrent networks for studying temporal structure in the experience loop.

8. Research Timeline

The project is organized into three distinct phases, each with a clear gate before the next begins. The structure is designed so that early work produces publishable results regardless of whether the full architectural synthesis succeeds.

Phase I: Literature and Foundations (Weeks 1–6)

The first six weeks are exclusively devoted to reading, understanding, and building the mathematical toolkit. No experiments. No code beyond setting up the computational pipeline. The goal is to arrive at Month 2 with genuine command of the relevant literature across three domains:

Weeks 1–2: Topology and Topological Data Analysis

- Work through Edelsbrunner & Harer, *Computational Topology*—focus on simplicial complexes, homology, and the persistence algorithm.
- Read Carlsson (2009), "Topology and Data"—understand the TDA paradigm and why persistence diagrams are the right summary object.
- Study Cohen-Steiner et al., "Stability of Persistence Diagrams"—this theorem is the mathematical foundation for using persistence as a robust measure.
- Install and learn Ripser/Gudhi/Giotto-TDA. Compute persistence diagrams on toy point clouds to build intuition.

Weeks 3–4: Cognition, Neuroscience, and Dynamical Systems

- Read Hebb, *The Organization of Behavior*—cell assemblies and the original structural theory of memory.
- Read Hopfield (1982)—associative memory as energy landscapes and attractors. This is the bridge between dynamical systems and memory.
- Read Gärdenfors, *Conceptual Spaces*—the strongest existing attempt to model concepts geometrically. Understand what it achieves and where topology goes further.
- Survey Kelso, *Dynamic Patterns*—cognition as nonlinear dynamical systems.

Weeks 5–6: Modern AI Architectures and Bridging Literature

- Read Vaswani et al. (Transformers), Ha & Schmidhuber (World Models), Gu et al. (S4/Mamba)—understand exactly what mathematical objects each operates on and what each optimizes.
- Read Papyan et al., "Neural Collapse"—emergent geometric structure in trained networks. This is the closest existing work to what we are proposing.
- Read Hofer et al., "Topological Regularization for Deep Learning" (NeurIPS 2019)—early work on topological loss functions.
- Finalize annotated bibliography. Identify the precise gap our work fills.

Phase I Deliverable: Annotated bibliography; working TDA computational pipeline; clear articulation of the gap between existing work and our hypothesis.

Phase II: Hypothesis Formalization and Function-by-Function Testing (Months 2–4)

With the literature internalized, Phase II formalizes the hypothesis and tests each cognitive function independently. Each function gets its own sub-experiment. The key discipline is: *one function at a time, with a clear pass/fail criterion before moving on.*

Month 2: Formalize Hypotheses and Design Experiments

- Define the mathematical objects precisely: cognitive state space S, memory structure M, concept region C, learning operator L, understanding map U, experience loop E(t).
- Write formal predictions for H1–H4 with specific quantitative criteria (e.g., "persistence lifetime of dominant H_0 features increases by at least 2× between epoch 1 and convergence").
- Design experimental protocols for each cognitive function.

Month 3: Test Memory and Concept Formation

- **Memory (H1):** Train classifiers on MNIST/CIFAR-10. Extract hidden layer activations at every epoch. Compute persistence diagrams. Track whether persistent features emerge and stabilize as training progresses. Pass criterion: persistence lifetimes increase monotonically with accuracy.
- **Concept Formation (H2):** At convergence, compute 0-dimensional persistence (connected components). Count persistent components vs. number of classes. Measure class purity per component. Pass criterion: persistent component count matches class count within ±2; purity > 80%.
- Run attractor analysis on Hopfield networks as a toy model of memory persistence. Does the number of stable attractors correspond to stored patterns?

Month 4: Test Learning Dynamics and Generalization

- **Learning (H4):** Compute Wasserstein distance between consecutive persistence diagrams throughout training. Plot topological learning rate vs. loss curve. Look for phase transitions—epochs where topology changes sharply but loss changes smoothly, or vice versa. Pass criterion: topological learning rate is not a monotonic function of loss.
- **Generalization vs. Overfitting (H3):** Train regularized and unregularized networks on the same data. Compare Betti numbers and total persistence at convergence. Pass criterion: overfitting network has strictly higher topological complexity.
- Compare across architectures (CNN vs. MLP vs. ResNet) to check whether topological signatures are architecture-independent.

Phase II Deliverable: Complete experimental results for all four hypotheses. For each cognitive function (memory, concepts, learning, generalization): a clear pass or fail against pre-registered criteria. If at least H1 and H2 pass, proceed to Phase III.

Phase III: Full Model and Cognitive Loop (Months 5–6)

Phase III is conditional: it proceeds only if the function-by-function tests in Phase II succeed. If they do, the final two months are devoted to synthesizing the individual results into a unified model and connecting it to the cognitive loop architecture.

Month 5: Build the Unified Model

- **Integrate the functions.** Build a single experimental system that demonstrates memory, concept formation, and learning dynamics together—not as separate experiments but as one evolving topological state space. Use a recurrent or continual learning setup where the network sees new classes over time and must retain old ones.
- **Implement the experience mechanism.** Model the Recursive Integrative Experience Loop as a recurrent dynamical process. At each timestep: integrate new input into the state space, recall from persistent memory structures, update the state graph, and consolidate. Track whether global connectivity (H_0 = 1 connected component) is maintained throughout—the topological signature of continuous integrated experience.
- **Topology-aware training.** Implement a topological regularization term: a loss component that penalizes unnecessary topological complexity (excess Betti numbers) during training. Compare topology-regularized networks against standard networks on both accuracy and structural metrics. If topology-regularized networks generalize better with simpler structure, this is strong evidence for the framework.
- **Connect to BrAIn architecture.** Map the experimental results onto the BrAIn cognitive loop: the state graph is the topological state space, the three memory layers are the persistent structures, the body map expansion is state space restructuring, and the cognitive loop's sense–think–gate–act–remember cycle is the experience mechanism operating on topological structure.

Month 6: Write the Paper

- **Paper 1** (submit within 6 months): "Persistent Topological Structure in Neural Representation Spaces During Learning." Contains: the four hypotheses, the experimental results, the topological learning rate as a novel diagnostic, and the connection between topological complexity and generalization.
- **Paper 2 outline** (begin drafting): "Topological Cognition: A Structural Framework for Memory, Learning, and Understanding in Dynamical State Spaces." Contains: the full formal framework, the unified model from Month 5, the experience mechanism, and the architectural connection to BrAIn.

Phase III Deliverable: Submitted Paper 1. Unified model demonstrating topological cognition in a single system. Paper 2 outline with preliminary results. If the experience mechanism maintains global connectivity while demonstrating persistent memory and concept formation, the groundbreaking result is established.

Summary

Phase	Weeks	Focus	Gate
I	1–6	Literature review across topology, cognition, and AI. Build computational pipeline. Identify the gap.	Can we precisely state what topology adds that geometry does not?
II	7–16	Formalize hypotheses. Test each cognitive function independently: memory (H1), concepts (H2), generalization (H3), learning dynamics (H4).	Do at least H1 and H2 pass their pre-registered criteria?
III	17–24	Build unified model. Implement experience mechanism. Topology-aware training. Write and submit Paper 1. Outline Paper 2.	Does the integrated system demonstrate persistent memory, concept formation, and continuous experience simultaneously?

Table 5. Three-phase research plan with explicit gates.

9. Required Reading

9.1 Topology and Topological Data Analysis

1. H. Edelsbrunner and J. Harer, *Computational Topology: An Introduction* (textbook).

2. G. Carlsson, "Topology and Data," *Bulletin of the AMS*, 2009.

3. D. Cohen-Steiner, H. Edelsbrunner, and J. Harer, "Stability of Persistence Diagrams," *Discrete & Computational Geometry*, 2007.

4. L. Wasserman, "Topological Data Analysis," *Annual Review of Statistics*, 2018.

9.2 Cognition, Memory, and Brain Function

1. D. Hebb, *The Organization of Behavior* (1949)—cell assemblies, Hebbian learning.

2. E. Kandel et al., *Principles of Neural Science*—standard neuroscience reference.

3. J. Hopfield, "Neural Networks and Physical Systems with Emergent Collective Computational Abilities," *PNAS* 1982.

4. J.A. Scott Kelso, *Dynamic Patterns: The Self-Organization of Brain and Behavior*.

5. P. Gärdenfors, *Conceptual Spaces: The Geometry of Thought*.

9.3 Modern AI Architectures

• A. Vaswani et al., "Attention Is All You Need," *NeurIPS* 2017.

• D. Ha and J. Schmidhuber, "World Models," 2018.

• A. Gu et al., *Structured State Spaces for Sequence Modeling (S4)*, 2022.

• A. Gu and T. Dao, "Mamba: Linear-Time Sequence Modeling with Selective State Spaces," 2023.

9.4 Bridging Literature: Topology Meets Learning

• V. Papyan et al., "Neural Collapse," 2020—emergent geometric structure in trained networks.

• L. McInnes et al., "UMAP: Uniform Manifold Approximation and Projection," 2018.

• C. Hofer et al., "Topological Regularization for Deep Learning," *NeurIPS* 2019.

• D. Spivak, *Category Theory for the Sciences*.

10. Final Objective and Scope

The Scope (6 Months)

Within six months, the goal is to produce **Paper 1**: a rigorous experimental study demonstrating that persistent topological features emerge in neural network representation spaces during learning, that these features correlate with learned categories, that overfitting produces excess topological complexity, and that topological change provides a structural learning signal distinct from the loss curve. This paper will be submitted to a workshop or journal.

The Ambition (Beyond 6 Months)

The larger goal is to develop a topological framework for cognition that provides formal mathematical definitions of memory, concepts, learning, understanding, and continuous experience in terms of persistent structure in dynamical state spaces. This framework will connect to the BrAIn cognitive architecture as the engineering system where these mathematical ideas are instantiated: where the cognitive loop is the dynamical system, the memory layers are the persistent structures, the state graph is the topological space, and the body map's expansion into new devices is a concrete instance of state space restructuring.

This project is not an attempt to solve cognition. It is an attempt to determine whether topology is the right mathematical language for describing cognition—and to provide the first evidence.

Arjun Paradkar & Arham Shah · April 2026

"Current AI models compute over vectors. We hypothesize cognition depends on the evolution of structured topological spaces."